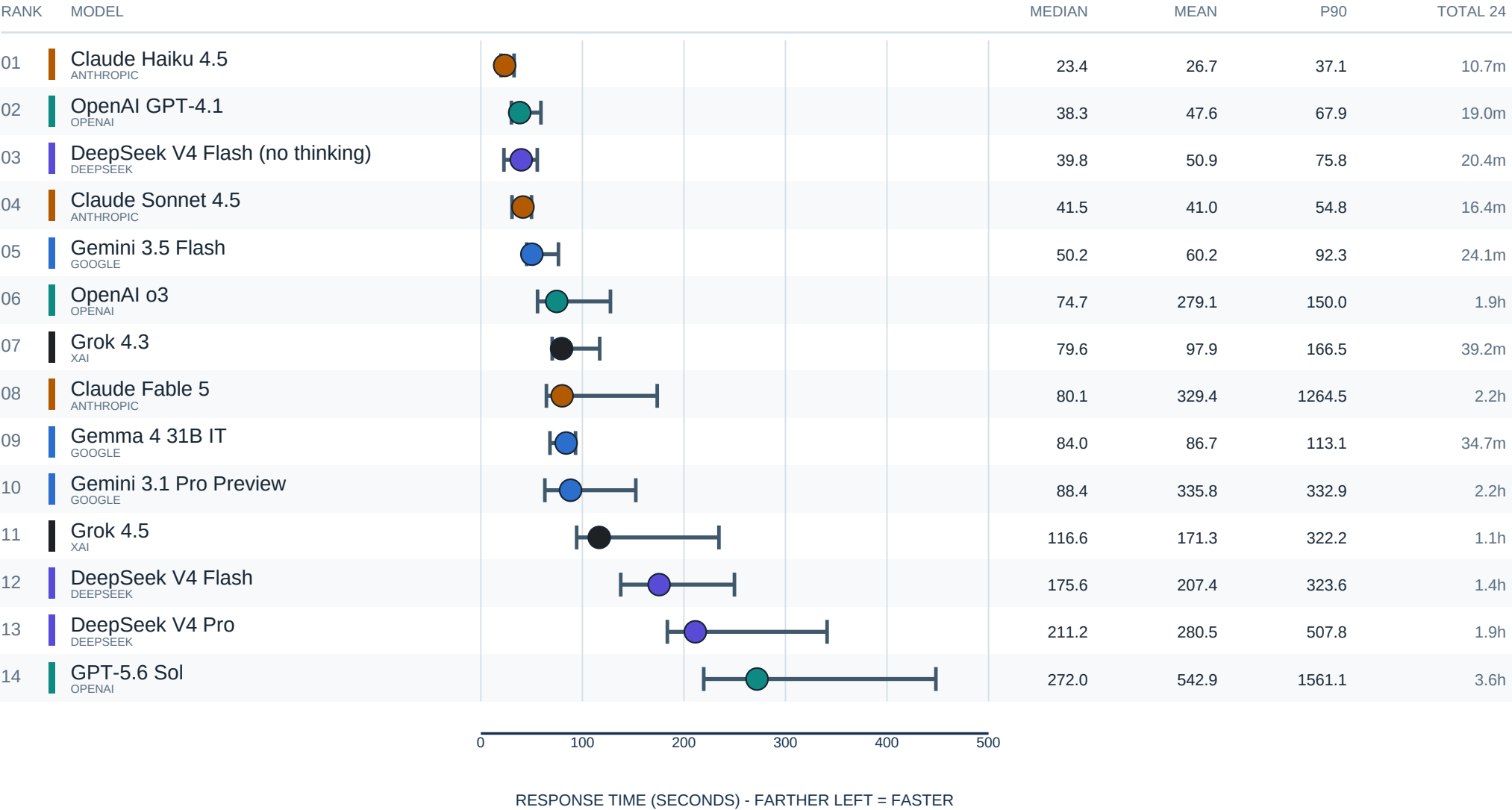


Typical response-time ranking

End-to-end time across 24 common exercises | ordered by increasing median

HOW TO READ  Dot = median; horizontal line with end caps = Q1-Q3 interval



Time excludes the local queue before dispatch; it includes the API call, generation, and any retries.

Where response times differ

Model-by-exercise heatmap | headers grouped by subject, exam session, and exercise

 Faster  Exercise median  Slower  retry

		ANALYSIS 3 - 2026																		PHYSICS 2 - 2026					
		Exam 1 A			Exam 1 B			Exam 2 A			Exam 2 B			Exam 3			Exam 4			Exam 3			Exam 4		
		E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3	E1	E2	E3
01	 Claude Haiku 4.5 ANTHROPIC	0.5	0.3	0.6	0.5	0.3	0.4	0.3	0.2	0.3	0.3	0.2	0.3	0.4	0.2	0.6	0.4	0.2	0.4	0.4	0.3	0.5	0.3	0.3	0.3
02	 OpenAI GPT-4.1 OPENAI	0.8	1.0	0.8	0.9	0.4	0.6	0.4	0.4	0.6	0.4	0.3	0.5	0.6	0.2	1.9	0.7	0.2	1.0	0.6	0.4	0.5	0.4	0.6	0.3
03	 DeepSeek V4 Flash (no thinking) DEEPSEEK	0.6	0.7	0.9	0.5	0.6	0.8	0.5	1.3	0.4	0.6	0.5	0.4	0.6	0.7	0.4	0.4	0.3	0.6	0.4	0.4	0.5	0.5	0.3	0.3
04	 Claude Sonnet 4.5 ANTHROPIC	0.8	0.5	0.6	0.7	0.8	0.6	0.4	0.2	0.6	0.4	0.3	0.7	0.5	0.3	0.7	0.5	0.3	0.8	0.6	0.6	0.7	0.4	0.4	0.4
05	 Gemini 3.5 Flash GOOGLE	0.7	0.5	0.6	0.7	0.7	0.6	0.6	0.6	0.8	0.6	0.5	0.6	0.9	0.7	1.0	1.0	0.6	0.9	1.1	0.6	1.0	0.8	0.7	0.8
06	 OpenAI o3 OPENAI	0.7	1.6	1.8	1.0	1.2	1.7	1.2	27.2	1.1	0.8	0.9	1.0	0.8	1.4	0.6	0.8	0.9	0.8	0.7	1.1	1.4	1.1	0.8	0.7
07	 Grok 4.3 XAI	1.1	1.0	1.0	1.0	1.0	0.8	1.3	1.0	0.9	1.7	1.0	1.5	1.0	1.3	1.0	1.2	1.1	1.0	1.0	1.3	1.1	1.4	1.2	2.0
08	 Claude Fable 5 ANTHROPIC	1.0	0.9	1.0	1.0	1.0	1.1	1.0	1.0	1.1	1.0	1.1	1.0	16.6	10.0	1.3	21.9	4.1	16.6	0.8	0.9	0.9	0.8	0.6	1.2
09	 Gemma 4 31B IT GOOGLE	1.1	1.1	1.2	1.0	1.5	1.4	1.0	1.0	1.4	1.0	1.0	1.2	1.0	0.6	1.0	1.0	0.7	1.4	1.0	0.9	1.1	0.9	1.3	0.7
10	 Gemini 3.1 Pro Preview GOOGLE	1.0	0.8	0.9	0.8	0.9	0.9	0.9	0.7	0.9	24.9	20.1	0.9	3.4	0.6	1.7	0.8	1.5	6.6	5.8	1.4	1.0	1.1	1.5	1.4
11	 Grok 4.5 XAI	1.4	1.4	1.8	1.3	1.3	1.6	1.3	3.1	4.4	2.8	2.1	1.9	1.3	1.7	1.0	1.2	1.7	1.3	1.1	1.9	1.0	1.4	2.2	6.6
12	 DeepSeek V4 Flash DEEPSEEK	2.5	2.2	2.8	2.6	2.0	2.5	3.2	1.9	2.1	3.3	2.1	2.2	5.9	2.0	3.3	1.6	1.3	2.2	2.5	2.1	1.3	1.9	2.2	3.0
13	 DeepSeek V4 Pro DEEPSEEK	3.0	3.0	2.7	3.0	2.1	3.0	2.4	2.7	3.1	3.8	2.8	2.1	3.4	5.6	3.2	2.5	1.3	0.9	1.0	2.6	1.3	7.1	3.3	10.0
14	 GPT-5.6 Sol OPENAI	3.4	2.8	3.6	3.5	3.0	3.2	3.6	2.1	1.8	3.1	2.1	1.8	13.1	6.0	6.9	19.4	15.3	48.6	3.3	4.0	4.6	3.2	2.6	5.3

0.5 means half the exercise's typical time; 2.0 means twice as long. Normalization controls exercise difficulty.

Common panel: 24 exercises (18 Analysis 3, 6 Physics 2), 14 models, 336 responses | 4 retry rows

How mean response time is formed

Token volume and effective time intensity jointly explain observed response time



Seconds per 1K is an operational ratio: it includes latency, generation, and retries, not pure provider throughput.

Response time vs accuracy

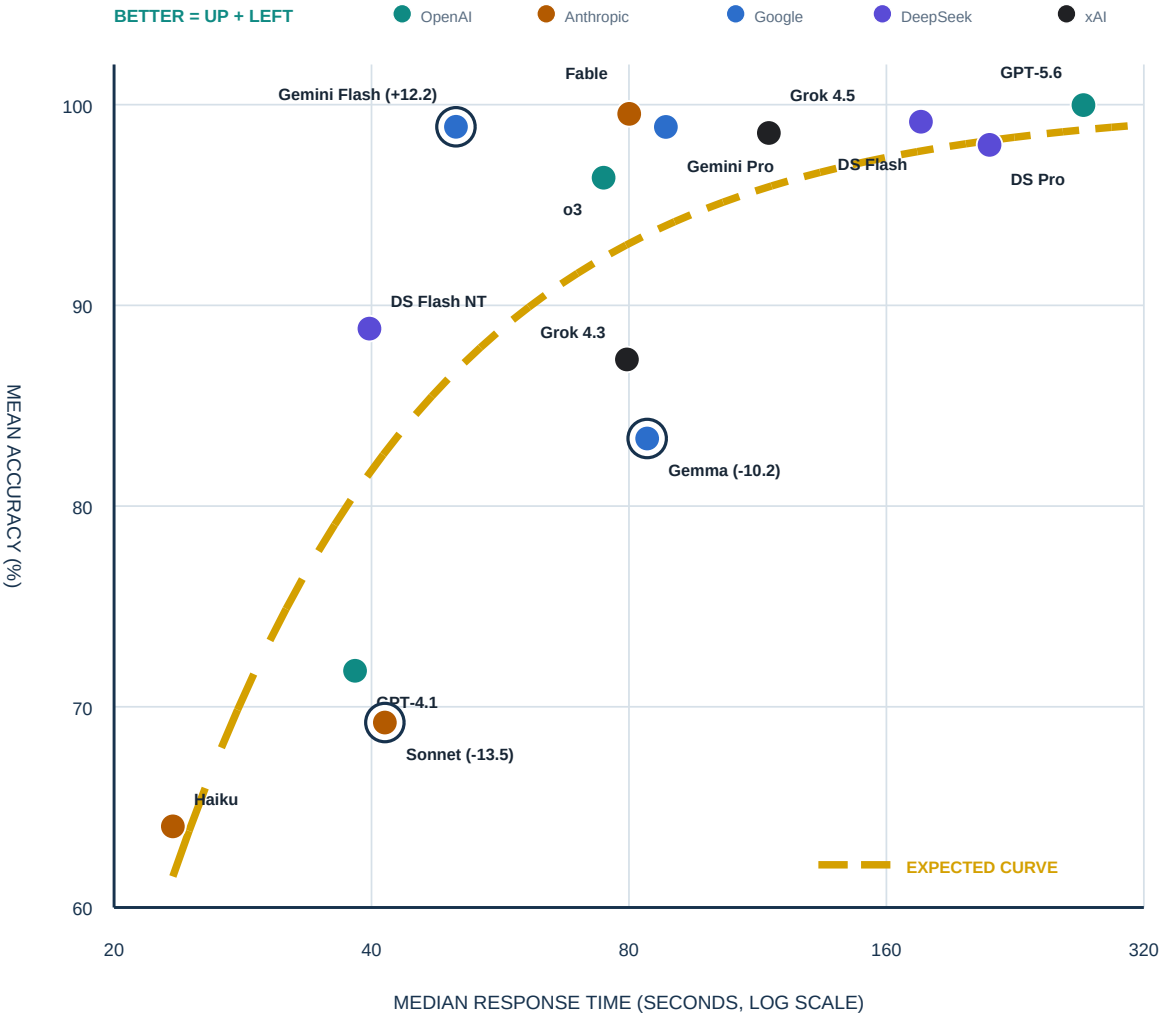
OPTION 4

Correlation and model deviations | 14 models x 24 common exercises

KEY RESULT Longer response time is associated with higher accuracy, but the gains flatten near the accuracy ceiling.

Pearson r (log time) = 0.74 | rank correlation ρ = 0.71 | bounded-curve R-squared = 0.67

1 OVERALL RELATIONSHIP



2 WHO BEATS OR MISSES THE TREND?

ACCURACY GAP (POINTS)

BELOW EXPECTED ABOVE EXPECTED



HOW TO READ: right of zero = more accurate than expected; left of zero = less accurate than expected. Rings mark the 3 largest gaps.